# Porcentômetro

Completeness Assessment System

Catálogo Taxonômico da Fauna do Brasil (CTFB)

---

User Manual — English

Version 2.1 · April 2026

**Access the application at:**

**https://comfortable-emotion-production-e366.up.railway.app/**

The **Porcentômetro** is a web tool developed for the *Catálogo Taxonômico da Fauna do Brasil* (CTFB). It provides two complementary analyses: (1) **Data Completeness** — how thoroughly mandatory fields are filled in for catalogued species; and (2) **Species-List Completeness** — a comparison of CTFB records against the Global Biodiversity Information Facility (GBIF) to identify species known from Brazil that are absent from the catalogue.

# 1. Accessing the System

**Open the application in any modern browser:**

**https://comfortable-emotion-production-e366.up.railway.app/**

You will be presented with a login page. Enter your username and password. If this is your first login after being added by an administrator, you will be prompted to create your own password before continuing.

## 1.1 Authorization

Each user is associated with one or more taxa listed in the researchers spreadsheet (*pesquisadores.xlsx*). You may analyse any taxon within your area of responsibility and all taxonomically subordinate groups. Users assigned to *Animalia* have access to the entire catalogue. Administrators may analyse any taxon.

## 1.2 First Access — Setting Your Password

Users added by an administrator via the bulk-import tool do **not** have a password assigned yet. On first login the system detects this and redirects you to a password-creation screen. Follow the steps below:

| | |
|---|---|
| **1** | Open the URL and type your username. Leave the password field **empty** and click *Login* (any value is ignored at this stage). |
| **2** | The system redirects you to the **Set Password** screen with a welcome message. |
| **3** | Enter a new password of **at least 6 characters**, retype it, then click *Set password and continue*. |
| **4** | If the entries match and are long enough, you are immediately logged in and taken to the main page. |
| **5** | From this point on, use your username and the new password for every subsequent login. |

> *If the passwords do not match or are shorter than 6 characters, an error is shown on the same screen and you may try again. No data is lost.*

## 1.3 Forgotten Password

If you forget your password, you cannot reset it yourself. Two recovery options:

- **Contact your administrator** — the administrator clicks *Reset password* next to your account in the Admin Dashboard. This clears your password and sets the "must set password" flag, so your next login follows the first-access flow.
- **Write to support** — e-mail `faunadobrasilctfb@gmail.com` with your username. The team will ask an administrator to reset your account.

## 1.4 Main Page

After login the main page shows two analysis-type cards. Click one, enter a taxon name in the text field (or click a chip from your authorised taxa list to fill it automatically), then press the action button.

# 2. Data Completeness Analysis

This analysis evaluates the fraction of mandatory fields filled in for each accepted species (*taxonomicstatus* = *NOME_ACEITO*, *taxonrank* = *ESPECIE* or *SUBESPECIE*) within the requested taxon. **Subspecies rule:** when a species has one or more accepted subspecies, the species-level row is excluded and only its subspecies are used as analysis units. Results are aggregated by family, order, class, sub-class, and phylum. When the input taxon is a family, an additional breakdown by genus is produced.

## 2.1 The 15 Evaluated Fields

The following fields are evaluated for every species (subject to ecological exclusion rules in §2.2):

| Field | Description |
|---|---|
| `bibliographicreference` | Primary bibliographic reference for the taxon. |
| `country` | Country of occurrence (Brazil = BR). |
| `establishmentmeans` | Native, invasive, introduced, etc. |
| `endemicbrazil` | Whether the species is endemic to Brazil. |
| `environment` | Environment type: epicontinental, marinho, or both. |
| `locality` | Occurrence locality description. |
| `epicontinentaldomain` | Phytogeographic domain (freshwater / terrestrial species). |
| `marinedomain` | Marine biogeographic province (marine species). |
| `lifeform` | Life-form / feeding guild. |
| `habitat` | Habitat description. |
| `vernacularname` | Common name in Portuguese. |
| `vernacularnamelanguage` | Language of the common name. |
| `vernacularnamelocality` | Region to which the common name applies. |
| `animalhostnames` | Animal host species (see §2.2 for exclusion rules). |
| `vegetalhostnames` | Plant host species (see §2.2 for exclusion rules). |

## 2.2 Ecological Exclusion Rules

Certain fields are excluded from both the numerator and the denominator when they are biologically inapplicable. The applicable denominator therefore varies per species. Lifeform detection uses substring matching to handle compound values (e.g. `herbivoro;vida livre individual`).

### Domain and country fields

| Field excluded | Condition | Rationale |
|---|---|---|

| | | |
|---|---|---|
| `marinedomain` | environment = *epicontinental* | A freshwater/terrestrial species has no marine biogeographic province. |
| `epicontinentaldomain` | environment = *marinho* | A strictly marine species has no continental phytogeographic domain. |
| one of the two domain fields | environment = *vazio* or blank | Always exactly ONE domain slot per species. For unknown environment, the filled domain is kept; if both or neither are filled, *marinedomain* is dropped by default. Additionally, *environment* itself is treated as not filled when its value is "vazio". |
| `country` | endemicbrazil = *SIM* | Species endemic to Brazil always occur in Brazil — the field adds no information. |

### Host fields

The treatment of `animalhostnames` and `vegetalhostnames` depends on the `lifeform` value (substring match applied to handle compound lifeform strings):

| lifeform value | Treatment |
|---|---|
| contains *endoparasito* or *ectoparasito* | Both columns are merged into **one combined host slot**. The slot counts as filled if *either* column is non-empty; blank if both are empty. The denominator gains 1 (not 2). |
| contains *herbivoro* (non-parasite) | `animalhostnames` is excluded (herbivores feed on plants). `vegetalhostnames` is evaluated normally. |
| contains *vida livre individual* (non-parasite) | Both host columns are excluded — free-living species do not have hosts. |
| blank / not filled | Both host columns are excluded — lifeform unknown, host relationship undefined. |
| any other value (e.g. *predador, colonial, sessil*) | Both `animalhostnames` and `vegetalhostnames` are **included normally** in the denominator. |

*Examples: (1) A freshwater beetle (environment = epicontinental) is scored over 14 fields — marinedomain removed. (2) An endoparasite is scored over 14 fields — the two host columns are replaced by one combined slot. (3) A free-living predator (lifeform = predador) is scored over 15 fields — both host columns count normally. (4) A species with lifeform = vida livre individual is scored over 13 fields — both host columns excluded. (5) A Brazilian endemic (endemicbrazil = SIM) is scored over one fewer field — country removed.*

## 2.3 Completeness Formula

For each species / subspecies the score is:

```
completeness (%) = (filled applicable fields) / (applicable fields) × 100
```

Results are aggregated per taxonomic group as mean, minimum, and maximum completeness across all units in that group. **Amplitude** (max – min) measures within-group heterogeneity.

## 2.4 Subspecies Rule

When a species has at least one accepted subspecies (*taxonrank* = *SUBESPECIE, taxonomicstatus* = *NOME_ACEITO*), the species-level row is **excluded** from the analysis and its accepted subspecies are used as the analysis units instead. This ensures that the completeness is measured at the finest accepted resolution available for each lineage.

## 2.5 Researcher Attribution

Each aggregated row lists the researcher (*nome_completo*) responsible for that group. The lookup starts at the group's own rank and climbs the hierarchy until a responsible researcher is found. When resolved above the row's rank, the cell is annotated with "(via <rank>)" to make delegation transparent.

## 2.6 Output Files

Results are packaged in a ZIP file containing:

| Sheet / File | Contents |
|---|---|
| `completeness_per_species.xlsx` | One row per species / subspecies: taxon_analisado, taxonomic context, taxonrank, completeness (%), filled_count, total_applicable, pesquisador. |
| `completeness_by_genus.xlsx` | Aggregation by genus (only when the input taxon is a family): n_spp_ctfb, mean/min/max completeness, amplitude, pesquisador. |
| `completeness_by_family.xlsx` | Aggregation by family: same columns as above. |
| `completeness_by_order.xlsx` | Aggregation by order. |
| `completeness_by_class.xlsx` | Aggregation by class. |
| `completeness_by_subclass.xlsx` | Aggregation by subclass (omitted if no subclass data). |
| `completeness_by_phylum.xlsx` | Aggregation by phylum (omitted for single-phylum queries). |
| `completeness_histograms_genus.pdf` | Horizontal bar chart of mean completeness by genus (only when input is a family), with min–max error bars. |
| `completeness_histograms_family.pdf` | Horizontal bar chart by family. The highest-mean bar is coloured red. |
| `completeness_histograms_order.pdf` | Same chart by order. |

| | |
|---|---|
| `completeness_histograms_class.pdf` | Same chart by class. |
| `completeness_histograms_subclass.pdf` | Same chart by subclass. |
| `completeness_histograms_phylum.pdf` | Same chart by phylum. |

# 3. Species-List Completeness (GBIF)

This analysis compares the CTFB accepted species list against species recorded in Brazil on GBIF, identifying species that occur in Brazil but are absent from the catalogue. The job runs in the background; the status page auto-refreshes every 2 seconds.

## 3.1 How It Works — Four Phases

### Phase 1 — Taxon resolution

The taxon name is resolved to a GBIF internal identifier (*usageKey*) via the GBIF Species Match API (api.gbif.org/v1/species/match). An error is returned if the name is not recognised.

### Phase 2 — Species key collection

The GBIF Occurrence Search API is queried with filters country=BR and taxonKey=<resolved key>, using facets on speciesKey to retrieve all species with at least one occurrence record in Brazil. Pagination is applied automatically for taxa with more than 50 000 species.

### Phase 3 — Name resolution

Each species key is resolved to a canonical binomial (genus + specific epithet) and its higher taxonomy (family, order, class, phylum) via the GBIF Species API. Up to 20 parallel requests run simultaneously. Fossil / extinct species (*extinct* = *true* in GBIF) are excluded at this stage.

### Phase 4 — Comparison with CTFB

CTFB species with *taxonomicstatus* = *NOME_ACEITO* and *taxonrank* = *ESPECIE* are extracted as lowercase canonical binomials. Each GBIF binomial is classified as: **present** (matched to an accepted name), **synonym** (matched to a synonym in CTFB), or **absent** (not found in any form).

## 3.2 Matching and Caveats

Matching is performed on the **lowercase canonical binomial**. Key caveats:

- Species with identical binomials in GBIF and CTFB are counted as present, regardless of author or year.
- A GBIF name matched to a CTFB synonym (non-NOME_ACEITO record with the same binomial) is classified as *Sinonimos_no_CTFB*, not as absent.
- Nomenclatural discrepancies — genus synonymy, spelling variants, different taxonomic treatments — cause a species to appear "absent" even if represented under a different name in CTFB. Manual review is recommended.
- Infraspecific taxa (subspecies, varieties) are excluded from both sides of the comparison.
- Fossil taxa are excluded (GBIF *extinct* = *true*).

## 3.3 Output File

A single XLSX file with four sheets:

| Sheet / File | Contents |
|---|---|

| | |
|---|---|
| `Ausentes_no_CTFB` | GBIF species (Brazil, non-fossil) absent from CTFB in any form. Columns: phylum_gbif, classe_gbif, ordem_gbif, familia_gbif, nome_cientifico, gbif_species_key. Sorted phylum → class → order → family → name. |
| `Sinonimos_no_CTFB` | GBIF species whose binomial matches a synonym (non-NOME_ACEITO) record in CTFB. Same columns as Ausentes_no_CTFB. |
| `Resumo_por_Familia` | Summary per family: total GBIF species, absent count, synonym count, present count. Sorted by absent count descending. |
| `Presentes_no_CTFB` | GBIF species matched to a CTFB NOME_ACEITO record. Useful for coverage cross-checking. |

## 3.4 Expected Run Time

| Taxon scale | Approx. GBIF species | Estimated time |
|---|---|---|
| Small order (e.g. Nematomorpha) | < 200 | 10–30 seconds |
| Medium order (e.g. Araneae) | ~5 000 | 2–5 minutes |
| Large order (e.g. Coleoptera) | ~100 000 | 15–30 minutes |
| Full phylum (e.g. Arthropoda) | > 200 000 | 1–3 hours |

*You may close the tab and return later; the job continues server-side until completion or the next redeployment.*

# 4. Administration

Administrator accounts access the Admin Dashboard via the link at the bottom of the main page.

## 4.1 User Management

- **Bulk import** — paste a list of usernames from *pesquisadores.xlsx*. Each user is created with the "must set password" flag. Usernames not found in the spreadsheet are rejected.
- **Individual add** — create a single user with an immediately usable password.
- **Export CSV** — download a backup of all user accounts including hashed passwords. Commit this file (*users_backup.csv*) to the repository before every redeployment so that users are automatically restored on startup.
- **Restore CSV** — re-import users from a previously exported CSV. Existing accounts are skipped; hashed passwords are restored.
- **Reset password** — clears a user's password and sets the "must set password" flag. The user follows the first-access flow on next login.

***Automatic seed on startup:*** *at every startup the application reads* `users_backup.csv` *from the project root and inserts any user that does not yet exist in the database (existing accounts and passwords are never overwritten). Export and commit this file after adding or changing users to guarantee they survive redeployments.*

## 4.2 Database Persistence

| Backend | Persistence | How to activate |
|---|---|---|
| PostgreSQL | Permanent — survives all redeployments. | Set DATABASE_URL to a postgresql:// connection string (e.g. from Railway). |
| SQLite | Ephemeral — lost on each redeployment. | Default when DATABASE_URL is not set. A warning banner appears on the Admin Dashboard. |

## 4.3 Data Sources

Two Google Sheets are loaded on startup. If unreachable, the system falls back to local XLSX files in the *data/* directory.

| File | Contents |
|---|---|
| `pesquisadores.xlsx` | Researcher registry: login, nome_completo, táxons, institution. Drives authorisation and researcher attribution. |
| `planilha_extendida.xlsx` | Extended CTFB catalogue (Darwin Core + manual fields). Source for all completeness calculations. |

# 5. Interpreting Results

## 5.1 Data Completeness Charts

Each PDF shows a horizontal bar per taxon. Bar length = mean completeness (%). Error bars extend to the minimum and maximum within the group. The highest-mean bar is coloured red; all others are blue. Percentages are annotated to the right.

A group with 100% mean and zero amplitude is perfectly homogeneous. Wide amplitude signals that some species are well-documented while others lack many fields — indicating where curatorial effort should be directed.

## 5.2 Species Gap List

The *Ausentes_no_CTFB* sheet lists candidate species for inclusion. Before acting on any entry, consider:

- Nomenclatural differences — the species may already be in CTFB under a synonymous binomial (check *Sinonimos_no_CTFB* first).
- Taxonomic scope — some GBIF records may refer to marine species outside CTFB's current coverage.
- Record quality — some GBIF records are misidentified or poorly georeferenced. High-uncertainty records should be verified.
- Subspecific taxa — the GBIF comparison covers species level only.